

Worksheet: RLHF, Reasoning Models & Codex

CS 5090 · Software Engineering with Generative AI · Week 5 · collected for completion credit. Discussing with your neighbor is encouraged; write your own answer. Everything on this sheet is fair game for Midterm 1.

Name: _____

Part 1 · RLHF, PPO, DPO

warm-up 1.1 Put the three stages in order, and say in a few words what the model learns from in each: **RLHF**, **pretraining**, **SFT**.

recall 1.2 InstructGPT trains in three steps: (1) SFT on demonstrations, (2) a reward model, (3) PPO.
(a) In Step 2, human labelers *rank* several model responses instead of writing the ideal response. Why is that a good idea?
(b) The reward model starts as a copy of the Step-1 model. What is changed about its output?

true / false 1.3 Circle T or F.

PPO trains directly on the human preference pairs.	T F
PPO increases expected reward while limiting how much the policy changes at each step.	T F
DPO needs a separately trained reward model.	T F
SFT says "here is the correct answer, imitate it"; RL says "try an answer, I will tell you how good it is".	T F

Part 2 · Reasoning and test-time scaling

apply 2.1 The slides warn that *fluent reasoning* $\neq$ *correct reasoning*. Give a short coding example of what that looks like.

classify 2.2 Label each as **test-time scaling** or **train-time scaling**:

Technique	Label
Force the model to keep thinking by appending "Wait" (budget forcing, as in s1)	
Run RL with verifiable rewards (RLVR) using GRPO	
Sample many candidate solutions and check them	
SFT on chain-of-thought reasoning traces	

read the plot 2.3 The s1 plots show accuracy rising as average thinking time (tokens) grows. What does this show, and what is the price you pay for it?

Part 3 · RLVR and GRPO

work it out 3.1 RLVR gives reward 1 if the answer is verified correct, else 0. The model writes 4 solutions to a coding prompt; only solutions 1 and 3 pass the unit tests. Rewards: [__ , __ , __ , __]. Why is this kind of reward easier to get for code than for "write a helpful answer"?

compare **3.2** DeepSeek-R1-Zero was trained with GRPO instead of PPO. Which model does GRPO eliminate, and why does that matter?

Part 4 • Codex, HumanEval, pass@k

recall **4.1** A HumanEval problem contains three things. Name them. When is a generated solution counted as correct?

work it out **4.2** A model generates 3 samples per problem (✓ = passes the tests):

Problem	sample 1	sample 2	sample 3
P1	✓	✗	✗
P2	✗	✗	✗
P3	✗	✓	✗
P4	✓	✓	✓

pass@1 (count only sample 1) = _____ pass@3 = _____
Why is this an example of test-time (inference-time) scaling?

exit ticket **4.3** Codex code can "look plausible but be wrong". Name two things you do before merging AI-generated code.